higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

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This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1 1.1.1

- Volume flow rate is defined as the amount of fluid moved in a given length of time.
- Mass flow rate is defined as the product of volume flow rate and the density of the fluid.

1.1.2 $Q_v = Av$ ✓ and $Q_m = Q_v \rho$✓
Where : A = area,
v = velocity of flow
ρ = density of fluid✓

(2 ×x 2) (4)

1.2 1.2.1 It consists of a pair of ultrasonic transducers✓ mounted diagonally with respect to the fluid flow axis.✓ Each transducer consists of a transmitter-receiver pair,✓ with the transmitter emitting ultrasonic energy that travels across to the receiver on the opposite side of the pipe.✓ (4)

1.2.2 Fluid flowing in the pipe causes a time difference between the transit times of beams traveling upstream and downstream.✓ The liquid velocity slows the signal that travels against the flow✓ and increases the received frequency✓ while speeding up the signal that travels with the flow✓ and decreasing the received frequency.✓ The difference in the measured frequencies is used to calculate the transit time of the ultrasonic beams✓ and thus the liquid velocity. (6)

[14]

QUESTION 2

2.1 $m = \frac{d^2}{D^2}$

$= \frac{200^2}{300^2}$✓

$= 0.44$✓ (2)

2.2 From the graph, the discharge coefficient C is 0,606. (1)

2.3 $E = \frac{1}{\sqrt{1-m^2}}$

$= \frac{1}{\sqrt{1-0.4444^2}}$✓

$= 1,1163$✓ (2)

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2.4 The expansion ε = 1,✓since water is an incompressible liquid.✓ (2)

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2.5 N = C.m.E
= 0.606 × 0.4444 × 1.1163
= 0.3006✓

$$N = \frac{Q\sqrt{\rho}}{0.01252\,D^2\sqrt{h}}$$

$$Q = \frac{0.3006\,x\,0.01252\sqrt{204}\;\;x\,300^2}{\sqrt{1000}}$$ ✓

= 152,9857 m³/h✓

$$R_D = \frac{3{,}54.Q\rho}{\mu d}$$

$$R_d = \frac{3.54\,x\,152{,}9857\,x\,1000}{0.0078\;\;x\,200}$$ ✓

R_d= 347159.8577✓ (5)

2.6 Z_R =1,006✓ m = 0.4444 and R_d = 347159,8577
Z_D = 1.000 m = 0,4444 and D = 300 mm
$Z = Z_R \times Z_D$
= 1.006 × 1.000
= 1.006✓ (3)

2.7

$$Q = 0.01252\,xCxZxEx\,\varepsilon xd^2\,x\sqrt{\frac{h}{\rho}}$$

$$Q = 0.01252\,x\,0.606\,x\,1.006\,x\,1.1163\,x\,1\,x\,200^2\,x\sqrt{\frac{204}{1000}}$$ ✓✓

= 153.9327 m³/h✓ (3)
[18]

TOTAL SECTION A: 32

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SECTION B

QUESTION 3

3.1 3.1.1 Density is defined as mass of a unit volume of a substance.

3.1.2 The concentration of water vapor in air and other gasses is called humidity.

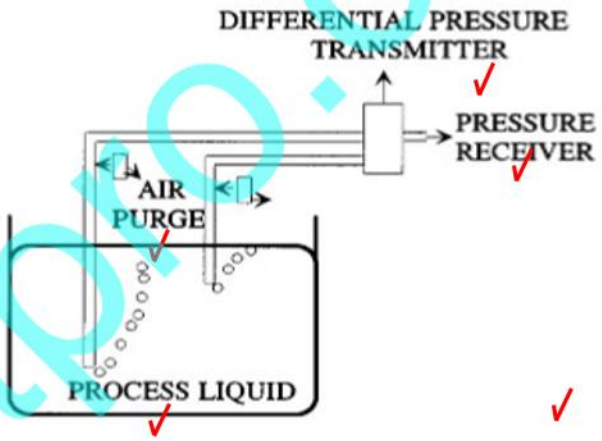
3.1.3 Viscosity is defined as the measure of a fluid's internal or intermolecular resistance to sheer stress.

3.1.4 Fluid which exhibits a constant ratio of shear stress to shear rate when subjected to a shear and undergoing constant deformation. Or when deformed by strain, the ratio between shear stress and shear rate is a constant value.

3.1.5 The temperature at which the saturation vapor pressure equals the partial pressure of water vapor in the atmosphere or other gas

(5 × 2) (10)

3.3 3.3.1

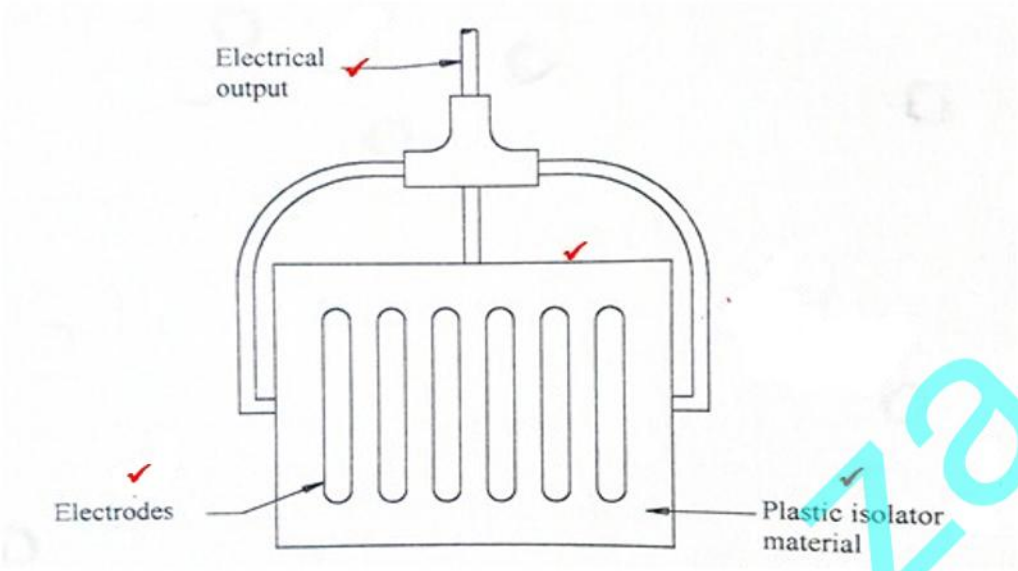


(4)

3.3.2 Constant air is supplied to both the tubes.✓ The pressure at the bottom of the longer tube will be higher than the pressure at the bottom of the shorter tube.✓ Thus more pressure is needed at the bottom of the longer tube for bubbles to form than at the bottom of the shorter tube.✓ When the bubbles become visible, the pressure put forth at the end of the tube by the liquid will be almost equal to the applied air pressure.✓ The difference in pressure will be directly proportional to the density.✓ (5)

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3.3



(4)

3.4
- The measuring element can be installed in a tank,open or closed✓ as long as the measuring tube is completely immersed in the fluid.✓
- During the filling phase, the piston at the bottom of the tube is automatically raised by an air-lifting mechanism✓ or by a motor-cam mechanism.
- As the piston is raised, a sample of the liquid to be measured is drawn in through openings in the sides of the tube,✓ and fills the measuring tube as the piston is wthidrawn.
- The filling and measuring cycle is controlled by periodic opening and closing of a valve.✓
- During the measuring phase, the piston assembly is allowed to fall by gravity,✓ expelling out of the tube through the same route that it entered.✓
- The time of all is a measure of viscosity.✓

(8)
[31]

QUESTION 4

4.1 When a substance combines with oxygen, electrons are transferred from that substance to the oxygen.✓ Such reactions involving exchange of electrons✓ are called oxidation-reduction reactions. (2)

4.2 Reduction = oxidation + ne,✓✓ where n is the number of electrons (2)

4.3
4.3.1 True
4.3.2 True
4.3.3 False
4.3.4 False

(4 × 1) (4)

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4.4 4.4.1 The pH of an aqueous solution is defined as the negative logarithm✓ of the active hydronium ion concentration,✓ (2)

4.4.2
$$pH = -\log\left[H_3O^+\right]$$
$$6 = -\log\left[H_3O^+\right]\checkmark$$
$$-6 = \log\left[H_3O^+\right]\checkmark$$
$$\left[H_3O^+\right] = 1\times10^{-6}\ \checkmark$$
(3)

4.4.3 The scale covers a range of values of 0,0 to 14.✓ (1)
[14]

QUESTION 5

5.1 To measure the process variable,✓ compared it with the set point.✓ It generates an error signal✓ and sends it to the correcting unit so that the error can be reduced✓ or to keep the process variable within the prescribe limits.✓ (4)

5.2 (1) Derivative control action response
(2) Integral control action response
(3) Proportional control action response (3)

5.3 5.3.1 Integral – sometimes called reset or floating control – is a control action causing the output signal to change over time ✓ at a rate proportional to the amount of error ✓ (the difference between PV and SP values).

5.3.2 Derivative – sometimes called rate or pre-act – is a control action causing the output signal to be offset ✓by an amount proportional to the rate at which the input is changing. ✓

OR

Derivative controller action responds to how quickly the input changes over time, biasing the output signal commensurate with that rate of input change.

(2 × 2) (4)

5.4 5.4.1
- All four bellows will have the same pressure.
- The motion pin will be on its true centre line.
- No control actions will be generated. (3)

5.4.2 The floating disc acting as a flapper will move towards the nozzle.✓ This should cause the nozzle feedback pressure to increase.✓ The output pressure will increase by 10 kPa✓ if the gain setting is direct .
(3)

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5.4.3 The floating disc acting as a flapper will move away from the nozzle.✓ This should cause the nozzle feedback pressure to decrease.✓ The output will decrease ±10 kPa✓ when the gain setting is indirect. (3)

5.4.4 $PB = \frac{100}{gain} = \frac{100}{1,0} = 100$ (1)

5.5 The measured variable is the amplitude of the signal being measured.✓ The error signal is the difference between the measured variable and the set point.✓ (2)
[23]

TOTAL SECTION B: 68
GRAND TOTAL: 100

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